

Inheritance

Example 1: C++ public Inheritance

```
// C++ program to demonstrate the working of public inheritance

#include <iostream>
using namespace std;

class Base {
private:
    int pvt = 1;

protected:
    int prot = 2;

public:
    int pub = 3;

    // function to access private member
    int getPVT() {
        return pvt;
    }
};

class PublicDerived : public Base {
public:
    // function to access protected member from Base
    int getProt() {
        return prot;
    }
};
```

Output:

Private = 1
Protected = 2
Public = 3
output:

Here, we have derived `PublicDerived` from `Base` in **public mode**.

As a result, in `PublicDerived`:

- `prot` is inherited as **protected**.
- `pub` and `getPVT()` are inherited as **public**.
- `pvt` is inaccessible since it is **private** in `Base`.

Example 2: C++ protected Inheritance

```
// C++ program to demonstrate the working of protected inheritance
```

```
#include <iostream>
using namespace std;

class Base {
private:
    int pvt = 1;

protected:
    int prot = 2;

public:
    int pub = 3;

    // function to access private member
    int getPVT() {
        return pvt;
    }
};

class ProtectedDerived : protected Base {
public:
    // function to access protected member from Base
    int getProt() {
        return prot;
    }

    // function to access public member from Base
    int getPub() {
        return pub;
    }
};

int main() {
    ProtectedDerived object1;
    cout << "Private cannot be accessed." << endl;
    cout << "Protected = " << object1.getProt() << endl;
    cout << "Public = " << object1.getPub() << endl;
    return 0;
}
```

Output:

```
Private cannot be accessed.
Protected = 2
Public = 3
```

Here, we have derived `ProtectedDerived` from `Base` in **protected mode**.

As a result, in `ProtectedDerived`:

- `prot`, `pub` and `getPVT()` are inherited as **protected**.
- `pvt` is inaccessible since it is **private** in `Base`.

As we know, **protected** members cannot be directly accessed from outside the class. As a result, we cannot use `getPVT()` from `ProtectedDerived`.

That is also why we need to create the `getPub()` function in `ProtectedDerived` in order to access the `pub` variable.

```
// Error: member "Base::getPVT()" is inaccessible
```

```
cout << "Private = " << object1.getPVT();
```

```
// Error: member "Base::pub" is inaccessible
```

```
cout << "Public = " << object1.pub;
```

Lab task:

Try to apply the private Inheritance on the previous example?